$$\text{Input} = [t_1, t_2, t_3 \ldots]^{1\times n}$$

$$E = \text{Emb Lookup}[t_1, t_2, t_3 \ldots]^{1\times n} \Longrightarrow E^{n\times d}$$

$$X = \begin{cases} E^{n\times d}\sqrt{d} + \sin\left(\frac{W_{pos}}{\delta_i}\right) & \text{if } 2(\text{mod } i) \equiv 0 \quad \text{(even)} \\ E^{n\times d}\sqrt{d} + \cos\left(\frac{W_{pos}}{\delta_i}\right) & \text{if } 2(\text{mod } i) = 1 \quad \text{(odd)} \end{cases}$$

$$Z^{(r)} = \text{Attention}(Q, K, V) = \left(\frac{Q_j K_j^T}{\sqrt{d_k}}\right) \odot M_{ij}(-\infty)^{j>i} \cdot V$$

splitting of X

$$\therefore \; Z^{(r)^{n\times d/k}} = \text{softmax}\left[\frac{X^{n\times d} W_Q^{d\times d/k} \cdot (X^{n\times d/k} W_K^{d\times d/k})^T}{\sqrt{d_k}} \odot M_{ij}^{n\times n}\right] \cdot X^{n\times d} W_V^{d\times d/k}$$

reintegration

$$\text{MHA}(H)^{n\times d} = \left[Z^{(1)} \,\|\, Z^{(2)} \,\|\, Z^{(3)} \ldots Z^{(k)}\right]^{n\times(k\times d/k) = n\times d} \cdot W_Z^{X\, d\times d}$$

Add back original X and layernorm;

$$x^{n\times d} = LN\left(\text{MHA}(H)^{n\times d} + X^{n\times d}\right)$$

$$= \alpha_x \odot \left[\frac{(\text{MHA}(H) + X) - \mu}{\sigma}\right] + \beta_x \qquad \alpha_x, \beta_x \text{ learnable}$$

<u>Feedforward</u>:

$$\text{FFN}(x)^{n\times d} = W_{F_2}^{d\times d}\left[\left(x^{n\times d} W_{F_1}^{d\times d} + b_1^{1\times d}\right) \text{GELU}\right] + b_2^{1\times d}$$

Add back previous step & layernorm

$$O_t^{n\times d} = LN\left(\overset{n\times d}{x} + FFN(x)^{n\times d}\right)$$

$$① \text{ —— } O_t^{n\times d} = \alpha_F \odot \left[\frac{(x + FFN(x)) - \mu}{\sigma}\right] + \beta_F$$

$$Y^n = \begin{cases} \text{Embed}(Y)^{n\times d}\sqrt{d} + \sin\left(\frac{w_{pos}}{\sigma_i}\right) & \text{if } 2(\text{mod } i) \equiv 0 \quad \text{(even)} \\ \text{Embed}(Y)^{n\times d}\sqrt{d} + \cos\left(\frac{w_{pos}}{\sigma_i}\right) & \text{if } 2(\text{mod } i) = 1 \quad \text{(odd)} \end{cases}$$

$$Z^{(r)^{n\times d_k}} = \text{softmax}\left\{\left[\frac{\overset{n\times d}{Y}\overset{d\times d_k}{W^Q} \cdot \overset{n\times d}{(Y}\overset{d\times d_k}{W^K)^T}}{\sqrt{d}}\right]^{n\times n} \odot M_{ij}(-\infty)^{j>i}\right\} \cdot \overset{n\times d}{Y}\overset{d\times d_k}{W^V}$$

$$MHA(Y)^{n\times d} = \left[Z^{(1)} \,||\, Z^{(2)} \,||\, \cdots Z^{(k)}\right]^{n\times k\times d_k = n\times d} \cdot W_Z^{Y\,d\times d}$$

Add back original X and layernorm;

$$Y_2^{n\times d} = LN\left(MHA(Y)^{n\times d} + Y^{n\times d}\right)$$

$$= \alpha_Y \odot \left[\frac{(MHA(Y) + Y) - \mu}{\sigma}\right] + \beta_Y$$

$$||$$

$$\Downarrow$$

$\boxed{\text{Cross attention}}$

$$Y_3^{n\times d} = LN\left(Y_2 + MHA_{cross}\left(Q = Y_2,\ K = \overbrace{O_t}^{\text{By } ①},\ V = O_t\right)\right)$$

$$= LN\left(Y_2^{n\times d} + \left[\left\{\left[\frac{Y_2^{n\times d} W_Q^{mx\ d\times d_k} \cdot O_t^{n\times d} W_K^{mx\ d\times k}}{\sqrt{d}} \odot M_{ij}\right] \cdot O_t^{n\times d} W_V^{mx\ d\times d_k}\right\} \Big\| \cdots\right]^{n\times h\times d_k = n\times d}\right)$$

$$O_F^{n\times d} = LN\left(Y_3^{n\times d} + \underbrace{\left[W_2^{d\times d}\left[\left(Y_3^{n\times d} \cdot W_1^{d\times d} + b_1^{1\times d}\right) GELU\right] + b_2^{1\times d}\right]}_{FFN}\right)$$

Predict using final output O_F

$$P(t_n \mid t_{n-1}, \ldots)^{n\times v} = \frac{e^{\left(O_F^{n\times d} \cdot W_{out}^{d\times v} + B_{out}^{1\times v}\right)_i}}{\sum e^{\left(O_F^{n\times d} \cdot W_{out}^{d\times v} + B_{out}^{1\times v}\right)}}$$

$$\mathcal{L} = -\sum_{i=1}^{N} \log P(\hat{y}_i, y_i)$$

$$Z^{n \times v} \xrightarrow[\text{one}]{\text{lets focus on}} Z^{1 \times v} \Rightarrow \frac{e^{z_i}}{\sum e^{z}} \Rightarrow P^{1 \times v}$$

one sentence of 'n' tokens

$$P(\hat{Y}) = P(y_1)^{1 \times v} \cdot P(y_2 \mid y_1)^{1 \times v} \cdots P(y_i \mid y_{i-1})^{1 \times v}$$

$$= \prod_{i=1}^{v} P(y_i \mid y_{i-1})$$

Take log

$$\ln P(\hat{Y}) = \ln\left(\prod_{i=1}^{v} P(y_i \mid y_{i-1})\right) = \sum_{i=1}^{v} \ln P(y_i \mid y_{i-1})$$

For one token

$$P(y_i \mid y_{i-1}) = \left\{ \left(\frac{e^{z_1}}{\sum e^{z}}\right), \left(\frac{e^{z_2}}{\sum e^{z}}\right), \ldots \right\}^{v}$$

for one logit

$$\ln P(z_j \mid y_i) = \ln\left(\frac{e^{z_j}}{\sum e^{z}}\right)$$

any jth logit out of v logits set in one y_i

$$= \ln e^{z_j} - \ln \sum e^{z}$$

$$\therefore \; -\ln P(z_j \mid y_i) = \ln \sum e^{z} - z_j \quad \text{———} \quad (1)$$

Now use $P_{i,t}^{n \times v \text{ (one hot)}}$ as ground truth in (1)

Multiply both sides

(2)

$$-\ln P(z_j \mid y_i) \cdot P_{i,j} = P_{i,j}\left[\ln \sum e^{z} - z_j\right] = \mathbb{1}_{j=t} \cdot \left(\ln \sum e^{z} - z_j\right)$$

$$\mathcal{L}_{z_j, y_i} = \mathbb{1}_{j=t} \cdot \left(\ln \sum e^{z} - z_j\right)$$ for one logit of one token

stack back logits to one token

$$\mathcal{L}_{y_i} = -\ln\left\{ \mathbb{1}_{j=t} \cdot P(z_1),\ \mathbb{1}_{j=t} \cdot P(z_2), \dots \mathbb{1}_{j=t} \cdot P(z_V) \right\}$$

$$= -P_{i,t} \ln P(z_{i,j})^{V}$$

stack back tokens

$$\mathcal{L} = -\sum_{i=1}^{V} \underbrace{P_{i,t}}_{\text{Label (one-hot)}} \cdot \ln \left(\underbrace{\left\{ P_i(z_1), P_i(z_2), \dots, P_i(z_V) \right\} \mid P_{i-1}}_{\text{model pred}^{n}} \right)$$

$$\mathcal{L}_i = \left(-\sum_{i=1}^{n} \ln P_{i,t} \right)^{1 \times 1}$$

Normalize scalar;

$$\boxed{\mathcal{L} = -\frac{1}{n} \sum_{i=1}^{n} \ln \cdot Y_{i,t_i}}$$

Next Goal : Find : $\frac{dh}{dz_i}$

Remember ① & ② ;